

Chapter 1: Computer Skills

Class 1 – Computer Fundamentals, Hardware Devices & Computer System

Learning Objectives

After completing this lesson, students will be able to:

- Understand the basic concepts of computers.
 - Identify the main parts of a computer system.
 - Differentiate between hardware and software.
 - Recognize internal and external hardware devices.
 - Explain the functions of common computer hardware.
 - Understand the importance and uses of computers in everyday life.
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Introduction

A **computer** is an electronic device that accepts data, processes it, stores information, and produces useful results. Computers help us perform many tasks quickly, accurately, and efficiently. Today, computers are used in schools, homes, hospitals, banks, offices, shops, and many other places.

Understanding computer fundamentals helps students develop digital skills that are useful for learning and everyday life.

What is a Computer?

A **computer** is an electronic machine that:

- Accepts data (Input)
- Processes the data
- Stores information
- Produces useful output

Simple Definition

Computer = Input → Process → Output → Storage

Examples of Computer Use

- Studying online
 - Playing educational games
 - Drawing pictures
 - Typing documents
 - Watching educational videos
 - Searching for information
 - Communicating with others
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Computer Fundamentals

Computer fundamentals are the basic concepts required to understand how a computer works.

They include:

- Computer System
 - Hardware
 - Software
 - Input Devices
 - Output Devices
 - Storage Devices
 - Operating System
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What is a Computer System?

A **computer system** is made up of different parts that work together to perform tasks.

Main Components of a Computer System

- Hardware
- Software
- User
- Data

Each component is important for the computer to function properly.

What is Hardware?

Hardware refers to all the physical parts of a computer that we can see and touch.

Examples

- Monitor
 - CPU (System Unit)
 - Keyboard
 - Mouse
 - Printer
 - Speakers
 - Scanner
 - Webcam
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Types of Hardware

1. Internal Hardware

Internal hardware is located **inside** the computer cabinet.

Examples

- Motherboard
- Processor (CPU)
- RAM (Memory)
- Hard Disk / SSD
- Power Supply Unit

Functions

- Processes data
 - Stores information
 - Controls computer operations
 - Supplies power to components
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2. External Hardware

External hardware is connected to the computer from the outside.

Examples

- Monitor
- Keyboard
- Mouse
- Printer
- Speakers
- Scanner
- Webcam
- Microphone

Functions

- Allows users to enter data
- Displays information
- Prints documents
- Produces sound
- Captures images and audio

Functions of Common Hardware Devices

Hardware Device	Function
Monitor	Displays text, images, and videos.
Keyboard	Used to type letters, numbers, and symbols.
Mouse	Controls the pointer and selects items.
CPU	Processes instructions and controls the computer.
Printer	Prints documents and pictures.
Scanner	Converts paper documents into digital form.
Speakers	Play sound, music, and videos.
Webcam	Captures photos and videos.
Microphone	Records voice and sound.

Input, Output & Storage (Overview)

Input Devices

Devices used to enter data into the computer.

Examples: Keyboard, Mouse, Scanner, Microphone

Output Devices

Devices used to display or produce information.

Examples: Monitor, Printer, Speakers

Storage Devices

Devices used to store data and files.

Examples: Hard Disk, SSD, Pen Drive, Memory Card

Uses of Computers in Daily Life

Computers are used in many places:

- Schools for learning
 - Homes for education and entertainment
 - Hospitals for patient records
 - Banks for financial transactions
 - Offices for business work
 - Shops for billing
 - Railway stations and airports for ticket booking
 - Libraries for managing books
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Advantages of Computers

- Fast processing
 - High accuracy
 - Large storage capacity
 - Easy communication
 - Supports online learning
 - Saves time and effort
 - Helps solve problems efficiently
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Basic Computer Terminology

Term	Meaning
Computer	An electronic device that processes data.
Hardware	Physical parts of a computer.
Software	Programs that tell the computer what to do.
CPU	The brain of the computer.
Monitor	Displays information on the screen.
Keyboard	Used for typing.
Mouse	Used to point, click, and select.
Storage	Saves data and files for future use.

Did You Know?

- The CPU is often called the "**Brain of the Computer.**"
 - Computers can perform millions of calculations in a second.
 - Modern computers are used in almost every profession.
 - Every computer system requires both **hardware and software** to work properly.
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Points to Remember

- A computer is an electronic device that processes data.
- A computer system consists of hardware, software, users, and data.
- Hardware includes all the physical parts of a computer.
- Internal hardware is inside the computer cabinet, while external hardware is connected from outside.
- Computers are widely used in education, healthcare, banking, business, and communication.
- Proper handling and care help computers work efficiently for a long time.